

Complete Nutrition & Fitness Guide 2026

Evidence-based guidelines for optimal health, performance, and body composition

1. Nutrition Fundamentals

1.1 Macronutrients

Macronutrients — proteins, carbohydrates, and fats — form the foundation of human nutrition. Protein provides 4 calories per gram and is essential for muscle repair, immune function, and enzyme production. Carbohydrates (4 cal/gram) are the primary fuel source for high-intensity exercise and brain function. Fats (9 cal/gram) support hormone production, vitamin absorption, and cell membrane integrity.

Recommended Daily Macronutrient Intake

Goal	Protein	Carbohydrates	Fat	Total Calories
Weight Loss	2.2g/kg	2-3g/kg	0.8g/kg	Deficit 300-500
Maintenance	1.6g/kg	4-5g/kg	1.0g/kg	TDEE
Muscle Gain	2.0g/kg	5-6g/kg	1.2g/kg	Surplus 200-300
Athletic Performance	1.8g/kg	6-8g/kg	1.5g/kg	TDEE + Training

1.2 Micronutrients

- Vitamin D: 2000-4000 IU/day — critical for bone health and immune function
- Magnesium: 400-420mg/day — supports sleep, muscle function, energy production
- Omega-3 fatty acids: 2-3g EPA+DHA/day — reduces inflammation, supports cardiovascular health
- Iron: 8mg/day (men), 18mg/day (women) — essential for oxygen transport
- Zinc: 11mg/day — immune function, testosterone production, wound healing

2. Training Principles

2.1 Resistance Training

For muscle hypertrophy, train each muscle group 2-3 times per week with 10-20 sets total volume. Rep ranges of 6-12 with 60-80% of 1RM are optimal for muscle growth. Progressive overload — gradually increasing weight, reps, or volume — is the primary driver of strength gains. Rest 60-120 seconds between sets for hypertrophy, 3-5 minutes for maximal strength.

2.2 Cardiovascular Training

The American Heart Association recommends 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity cardio per week. Zone 2 training (60-70% max HR) improves aerobic base and fat oxidation. HIIT (High-Intensity Interval Training) improves VO2 max and metabolic rate but requires adequate recovery.

3. Recovery & Sleep

Sleep is the most underrated performance enhancer. Adults require 7-9 hours per night. During sleep, growth hormone is released, muscle protein synthesis occurs, and cognitive function is restored. Chronic sleep deprivation (< 6 hours) reduces testosterone by 15%, increases cortisol, impairs glucose metabolism, and reduces reaction time equivalent to being legally drunk.

4. Sample Weekly Training Plan

Day	Training Focus	Duration	Intensity
Monday	Upper Body Push (Chest, Shoulders, Triceps)	60 min	High
Tuesday	Lower Body (Quads, Hamstrings, Calves)	60 min	High
Wednesday	Zone 2 Cardio + Mobility	45 min	Low
Thursday	Upper Body Pull (Back, Biceps)	60 min	High
Friday	Full Body Power + Core	60 min	Moderate
Saturday	Active Recovery / Sports	30-60 min	Low
Sunday	Complete Rest	—	—